

SIEM ALERT TRIAGE WORKFLOW

SOC Analyst Lab Project

Lab Date: 29 March 2026

Report Date: 1 April 2026

Project by: Mohammad Mubashir

Environment: Windows 11, Kali Linux

Tools Used: Kali Linux, Splunk, Windows 11, Hydra, Crowbar,
Event Viewer

1. INTRODUCTION

1.1 Project Objective

The main objective of this project is to simulate an RDP brute force attack and perform alert triage using Splunk. This project demonstrates how SOC analysts detect, investigate, and respond to security alerts in a SIEM environment.

1.2 Scope

This project covers:

- Simulating RDP brute force attack using Crowbar/Hydra
- Ingesting Windows Security Logs into Splunk
- Searching for failed login events (Event ID 4625)
- Creating alert rules for brute force detection
- Documenting incident response workflow

1.3 Tools Used

Tool	Purpose
Windows	Victim machine + Splunk indexer
Kali	Attacker Machine
Splunk	SIEM for log ingestion and alerting
Crowbar/Hydra	RDP brute force tools
Event Viewer	Windows log analysis

2. LAB SETUP

2.1 Lab Environment

Machine	Operating System	IP Address	Role
Attacker	Kali Linux	192.168.1.43	RDP brute force attack
Victim/SIEM	Windows 11	192.168.1.34	Splunk Indexer + Target

2.2 Network Setup

Both machines are connected through a private virtual network using VirtualBox Bridge Adapter. This ensures an isolated lab environment without external interference.

2.3 Splunk Configuration

Component	Configuration
Splunk Version	10.2.1
Receiver Port	9997
Web Port	8000
Log Source	Windows Security Logs (Event ID 4625)

3. GENERATE RDP BRUTE FORCE ATTACK

3.1 First Attempt – Hydra RDP Error

Command (Kali Linux): `hydra -l testuser -P /usr/share/wordlists/rockyou.txt rdp://192.168.1.34`

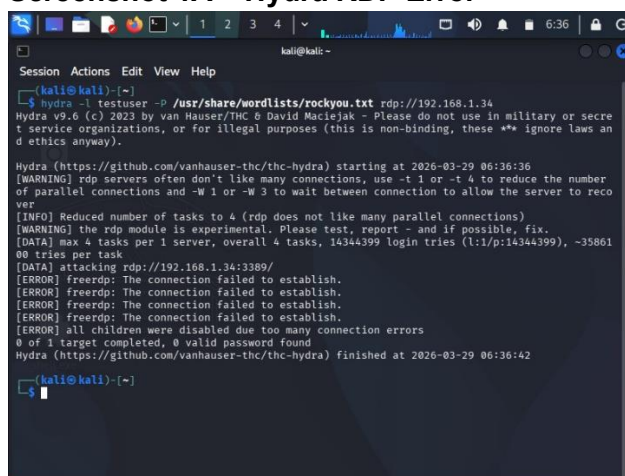
Error: [ERROR] freerdp: The connection failed to establish.

[ERROR] all children were disabled due too many connection errors

Reason:

Hydra's RDP module is experimental and unstable. It fails to maintain consistent connections with Windows RDP service.

Screenshot 1A – Hydra RDP Error



```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali:~$ hydra -l testuser -P /usr/share/wordlists/rockyou.txt rdp://192.168.1.34  
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or security service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-29 06:36:36  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~35861 00 tries per task  
[DATA] attacking rdp://192.168.1.34:3389/  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] all children were disabled due too many connection errors  
0 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-29 06:36:42  
kali@kali:~$
```

3.2 Solution – Alternative Tool (Crowbar)

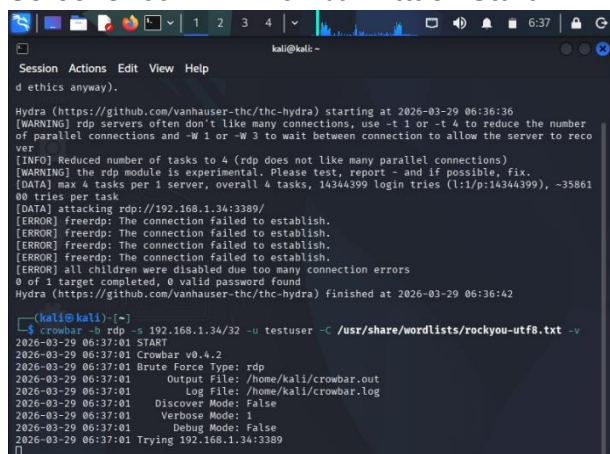
Command (Kali Linux):

`crowbar -b rdp -s 192.168.1.34/32 -u testuser -C /usr/share/wordlists/rockyou-utf8.txt -v`

Why Crowbar?

Crowbar is specifically designed for RDP brute force attacks and is more stable than Hydra.

Screenshot 1B – Crowbar Attack Start



```
kali@kali: ~  
Session Actions Edit View Help  
d ethics anyway).  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-29 06:36:36  
[WARNING] rdp servers often don't like many connections, use -t 1 or -t 4 to reduce the number of parallel connections and -W 1 or -W 3 to wait between connection to allow the server to recover  
[INFO] Reduced number of tasks to 4 (rdp does not like many parallel connections)  
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.  
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~35861 00 tries per task  
[DATA] attacking rdp://192.168.1.34:3389/  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] freerdp: The connection failed to establish.  
[ERROR] all children were disabled due too many connection errors  
0 of 1 target completed, 0 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-29 06:36:42  
kali@kali:~$ crowbar -b rdp -s 192.168.1.34/32 -u testuser -C /usr/share/wordlists/rockyou-utf8.txt -v  
2026-03-29 06:37:01 START  
2026-03-29 06:37:01 Crowbar v0.4.2  
2026-03-29 06:37:01 Brute Force Type: rdp  
2026-03-29 06:37:01 Output File: /home/kali/crowbar.out  
2026-03-29 06:37:01 Log File: /home/kali/crowbar.log  
2026-03-29 06:37:01 Discover Mode: False  
2026-03-29 06:37:01 Verbose Mode: 1  
2026-03-29 06:37:01 Debug Mode: False  
2026-03-29 06:37:01 Trying 192.168.1.34:3389  
[
```

3.3 Attack Execution

Command (Kali Linux):

```
crowbar -b rdp -s 192.168.1.34/32 -u testuser -C /usr/share/wordlists/rockyou-utf8.txt -v
```

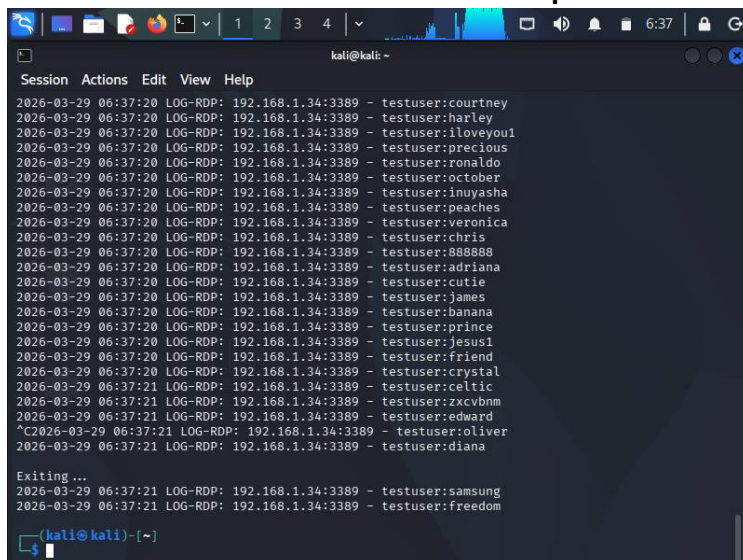
Output:

```
LOG-RDP: 192.168.1.34:3389 - testuser:123456
LOG-RDP: 192.168.1.34:3389 - testuser:password
LOG-RDP: 192.168.1.34:3389 - testuser:iloveyou
```

Observation:

The attack generated multiple failed login attempts but caused 100% CPU usage on the Kali machine. To prevent system slowdown, the attack was stopped using Ctrl + C.

Screenshot 1C – Crowbar Attack Attempts



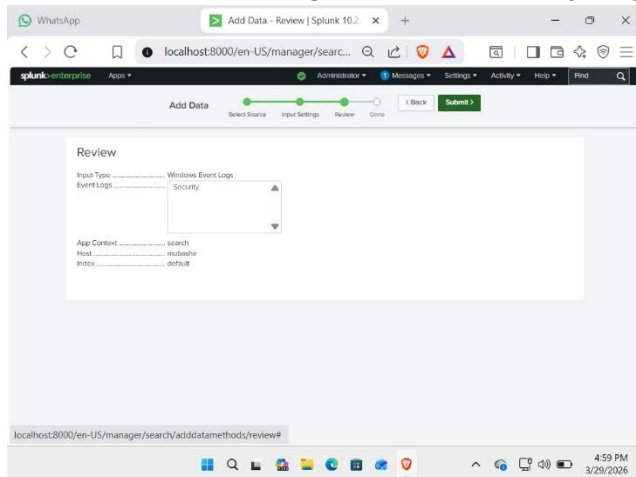
4. INGEST WINDOWS SECURITY LOGS

4.1 Add Windows Event Logs to Splunk

Steps:

1. Open Splunk Web: <http://192.168.1.34:8000>
2. Go to **Settings** → **Add Data**
3. Select **Monitor** → **Local Event Logs**
4. Choose **Security** from the list
5. Set **Source type** to WinEventLog:Security
6. Click **Review** → **Submit**

Screenshot 2 – Adding Windows Security Logs



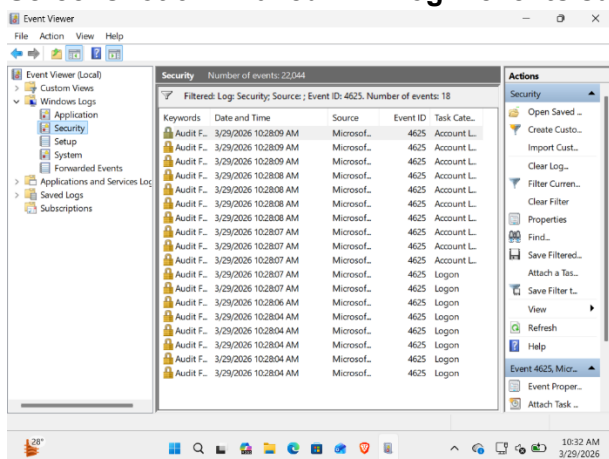
4.2 Verify Logs in Splunk

index=* sourcetype=WinEventLog:Security EventCode=4625

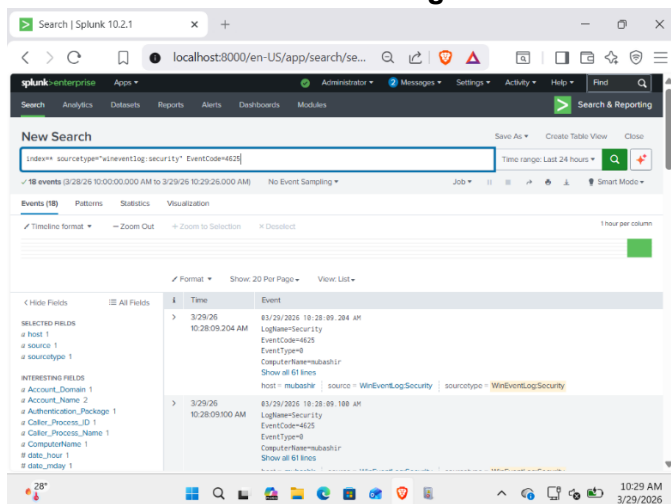
Purpose:

This search displays all failed login attempts (Event ID 4625) from Windows Security Logs.

Screenshot 3A - Failed RDP login events successfully ingested and visible in Splunk



Screenshot 3B - Failed RDP login events successfully ingested and visible in Splunk



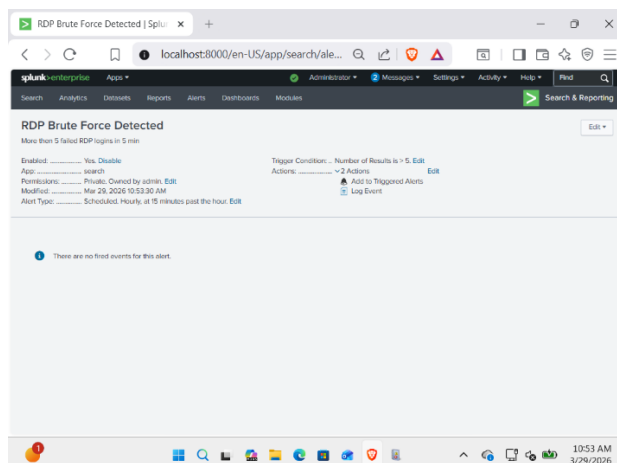
Field	Value
Event ID	4625
Source IP	192.168.1.43
Target User	testuser
Computer Name	Mubashir
Event Type	0
Status	0xc000006d (Wrong password)

5. SPLUNK ALERT CONFIGURATION

5.1 Create Alert Rule

Field	Value
Alert Name	RDP Brute Force Detected
Description	More than 5 failed RDP logins in 5 min
Status	Enable
Alert Type	Scheduled (Hourly, at 15 minutes past the hour)
Trigger Condition	Number of Results > 5
Actions	Add to Triggered Alerts, Log Event

Screenshot 4 – Alert Rule Created



5.2 Alert Status

Alert was configured and verified in Splunk. Due to early termination of the attack, the alert threshold was not reached during this session. In a real SOC environment, this alert would trigger automatically upon 5+ failed logins within the defined time window.

6. INCIDENT RESPONSE WORKFLOW

6.1 Incident Response Steps

Steps	Action	Owner
1	Alert Triggered	Splunk
2	Triage Alert	SOC L1
3	Verify False Positive	SOC L1
4	Escalate to L2	SOC L1
5	Block Attacker IP	Firewall Team
6	Document Findings	SOC L1

6.2 Escalation Matrix

Severity	Escalation Path	Response Time
Critical	Immediate to L2 + SOC Manager	< 15 mins
High	Escalate to L2	< 30 mins
Medium	Document + Monitor	< 1 hour
Low	Close as FP	N/A

6.3 Sample Ticket

Field	Value
Ticket ID	INC-001
Alert Name	RDP Brute Force Detected
Severity	High
Source IP	192.168.1.43
Target IP	192.168.1.34
Action Taken	IP blocked, user notified

7. SUMMARY & IOC DOCUMENTATION

7.1 Project Summary

This project simulated an RDP brute force attack from Kali Linux to Windows 11. Windows Security Logs were ingested into Splunk, and an alert rule was created to detect multiple failed login attempts (Event ID 4625). An incident response workflow was documented for SOC operations.

7.2 Attack Simulation Details

Parameter	Value
Attacker IP	192.168.1.43 (Kali)
Target IP	192.168.1.34 (Windows 11)
Attack Tool	Crowbar / Hydra
Target user	testuser
Password list	rockyou-utf8.txt
Attack status	Stopped early due to high CPU usage

7.3 Lessons Learned

- Hydra RDP module is unstable — Crowbar is better for RDP brute force
- Wordlist encoding issues can be fixed with iconv
- High CPU usage may require stopping attacks early
- Splunk alert rules require at least one action (e.g., Triggered Alerts)

7.4 IOCs Identified

IOC NO.	Type	Value	Description
IOC 01	Network (Source IP)	192.168.1.43	Attacker IP (Kali Linux)
IOC 02	Network (Target IP)	192.168.1.34	Victim IP (Windows 11)

IOC 03	Host (Event ID)	4625	Failed login attempt
IOC 04	Authentication	testuser	Targeted username
IOC 05	Logon Type	10	Confirms RDP based remote login attempt
IOC 06	Status Code	0xc000006d	Wrong password
IOC 07	Attack Tool	Crowbar / Hydra	Brute force tool used

7.5 IOCs Severity

IOC NO.	Severity	Justification
IOC 01	Critical	Attacker IP identified
IOC 02	High	Target IP exposed
IOC 03	Critical	Multiple failed logins indicate brute force
IOC 04	Medium	Username enumeration possible
IOC 05	High	RDP brute force attempt
IOC 06	Low	Generic error code
IOC 07	Medium	Tool signature may be detected

8. GITHUB REPOSITORY

<https://github.com/Mubashir-soc-01/SOC-Project>

9. LINKEDIN PROFILE

<https://www.linkedin.com/in/mohammad-mubashir-b729643b7/>

THANK YOU

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